

Chapter 67 — Male Reproductive Physiology

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HYPOTHALAMIC-PITUITARY-GONADAL AXIS

- The **hypothalamic-pituitary-gonadal (HPG) axis** governs four processes, in development and adulthood: **phenotypic sex development** in embryogenesis → **sexual maturation at puberty** → **testis endocrine function** (testosterone) → **testis exocrine function** (sperm).

Basic Endocrine Concepts

PEPTIDE VERSUS STEROID HORMONES

	Peptide	Steroid
Nature	Small secretory proteins	Derived from cholesterol ; lipophilic
Storage	Secretory granules	Not stored — secretion directly reflects production rate
Receptor	Cell-surface	Intracellular ; membrane-permeable
Signalling	Second messengers (cyclic adenosine monophosphate [cAMP], calcium, tyrosine kinase) → protein phosphorylation	Receptor translocates to nuclear DNA recognition sites → regulates gene transcription
Plasma	—	Bound to carrier proteins
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol

- Signalling is **hierarchically governed by a free-running hypothalamic pulse generator**; the **amplitude and frequency** of secretion determine downstream organ responsiveness.
- **Feedback control** is the principal regulatory mechanism — a hormone may regulate its own synthesis and action or another's; **negative feedback** minimises perturbation and maintains homeostasis.

Concept — *the axis is read by its rhythm as much as its level: pulse frequency and negative feedback, not absolute hormone quantity, set downstream responsiveness.*

Components of the Reproductive Axis

Hypothalamus

- **Integrative centre** of the axis: input from amygdala, thalamus, pons, retina, olfactory cortex and other areas; it is the **pulse generator** for cyclic pituitary secretion.
- Linked to the pituitary by **neuronal pathways and a portal vascular system**; the portal system bypasses the systemic circulation, delivering hypothalamic hormones directly to the anterior pituitary.
- **Gonadotrophin-releasing hormone (GnRH)**: **10-amino-acid peptide** from neurones of the **preoptic and arcuate nuclei**; its **only known function** is to stimulate LH and FSH secretion.
- Plasma **half-life about 5–7 minutes**; almost entirely **removed on first pass through the pituitary** (receptor internalisation or enzymatic degradation).
- GnRH output integrates stress, exercise and diet from higher centres, pituitary gonadotrophins and circulating gonadal hormones (Table 67.1).
- **Kallmann syndrome**: congenital hypogonadotrophic hypogonadism — **GnRH precursor neurones fail to migrate** → absent hypothalamic GnRH → delayed puberty or infertility from absent testosterone production.

- Three rhythms: **seasonal** (months; peaks in spring), **circadian** (higher testosterone in the early morning), **pulsatile** (peaks every 90–120 minutes).
- **Pulsatility is essential**: exogenous **GnRH agonists** (e.g. leuprolide acetate) **stop testicular testosterone production** by converting pituitary GnRH exposure from cyclic to constant.

TABLE 67.1 SUBSTANCES THAT MODULATE GONADOTROPHIN-RELEASING HORMONE (GNRH) SECRETION

Direction of feedback	Modulator class	Example
Negative	Opioids	β-Endorphin
	Peptide hormones	FSH, LH
	Sex steroids	Testosterone
Positive	Prostaglandins	Prostaglandin E ₂ (PGE ₂)
	Insulin	Insulin
	Kisspeptins	Kisspeptin (puberty)
	Leptins	Leptin
Variable	Catecholamines	Dopamine

- **What it is for**: the inhibitors are stress signals (opioids) and the axis's own end-products (gonadotrophins, testosterone); the stimulators are **metabolic and nutritional signals** (insulin, leptin) and the **pubertal trigger** (kisspeptin).
- **Trap**: catecholamines are **variable**, not simply inhibitory; kisspeptin and leptin are **positive**.

Anterior Pituitary

POSTERIOR VERSUS ANTERIOR PITUITARY

Posterior lobe (neurohypophysis)	Anterior lobe (adenohypophysis)
Oxytocin, vasopressin	LH, FSH, prolactin, adrenocorticotrophic hormone (ACTH), growth hormone (GH), thyroid-stimulating hormone (TSH)
Driven by neural stimuli	Regulated by blood-borne factors ; site of GnRH action

- Pituitary lies within the **bony sella turcica**.
- GnRH stimulates FSH and LH production and release by a **calcium flux-dependent** mechanism; gonadotroph sensitivity to GnRH **varies with age and hormonal status**.
- LH and FSH are **glycoproteins of two subunits, α and β**, each coded by a separate gene. The **α subunit is identical** between them and similar to other pituitary glycoprotein hormones; the **unique β subunit confers biological and immunological specificity**. Both subunits are required for activity.
- Their oligosaccharide side-chains (sialic acid residues) differ in content and **likely account for different plasma clearance rates**.
- Both act **only in the gonads**, via adenylate cyclase → intracellular cAMP.

LUTEINISING HORMONE VERSUS FOLLICLE-STIMULATING HORMONE

	LH	FSH
Pulses	8–16 per 24 h ; amplitude varies one- to threefold	About every 1.5 h ; amplitude varies by about a quarter
Relation to GnRH	Pulses closely reflect GnRH release	Relatively independent of GnRH (inhibin, activin)
Response to GnRH	Readily assessed	Harder to assess : smaller amplitude, longer serum half-life

	LH	FSH
Main feedback	Testosterone (androgens and oestrogens, negative)	Oestradiol and inhibin
Target	Leydig cells	Sertoli cells and spermatogonial membranes
Action	Steroidogenesis: mitochondrial cholesterol → pregnenolone → testosterone	Seminiferous tubule growth in development; essential for initiating spermatogenesis at puberty ; in the adult, maintains quantitatively normal spermatogenesis

- **Prolactin**: globular protein of **199 amino acids (23 kDa)**; milk synthesis in pregnancy and lactation; **no human mutations** known in the prolactin gene or its receptor.
- Male role less clear: may **increase Leydig cell LH receptors** and sustain high intratesticular testosterone; may potentiate androgen effects on accessory sex glands; normal levels may be needed for **libido**.
- **Low prolactin is not necessarily pathological**; **hyperprolactinaemia abolishes gonadotrophin pulsatility** by interfering with episodic GnRH release.
- ACTH, GH and TSH from the anterior pituitary can also significantly affect male reproduction.

Testis

- Virility and fertility need the **endocrine testis** (interstitial Leydig cells — steroidogenesis) and the **exocrine testis** (seminiferous tubules — spermatozoa) working together.
- Testosterone production is **about 5 mg/day**, secreted in a **damped, irregular, pulsatile (nyctohemeral)** pattern.

ACTIVE METABOLITES OF TESTOSTERONE

Metabolite	Enzyme	Note
Dihydrotestosterone (DHT) — the major androgen	5α-reductase	Much more potent than testosterone; conversion needed in most peripheral tissues, but not in testis or skeletal muscle
Oestradiol	Aromatase	Contributes to feedback, especially on FSH

- FSH acts primarily on **Sertoli cells**, inducing **androgen-binding protein (ABP)**, transferrin, lactate, ceruloplasmin, clusterin, plasminogen activator, prostaglandins and growth factors → tubule growth in development, sperm production from puberty.
- **Mouse FSH knockouts** suggest FSH is **not essential** for spermatogenesis — affected mice can be fertile.

INHIBIN VERSUS ACTIVIN

Inhibin	Activin
32-kDa protein from Sertoli cells	Testis protein, close homology to transforming growth factor-β (TGF-β)
Inhibits FSH release; production stimulated by FSH; negative feedback at pituitary or hypothalamus	Stimulates FSH secretion; receptors in many extragonadal tissues — possible growth-factor roles

- Steroid negative feedback on GnRH acts through **androgen receptors (ARs)** in hypothalamus and pituitary; **both testosterone and oestrogen** participate — mainly testosterone, lesser oestradiol.

STEROID FEEDBACK: TESTOSTERONE VERSUS OESTRADIOL

	Testosterone	Oestradiol
Main site	Hypothalamus	Pituitary
Predominant target	LH (primary regulator)	FSH , together with inhibin

Concept — the axis runs two loops: an LH loop between hypothalamus and Leydig cell governed by testosterone, and a partly GnRH-independent FSH loop between pituitary and Sertoli cell governed by oestradiol and inhibin — so LH reports Leydig function and FSH and inhibin report the tubules.

Development of the Hypothalamic-Pituitary-Gonadal Axis

- Sex determination is **genetic**. **SRY** (**sex-determining region Y**) on the **short arm of the Y chromosome** encodes a protein with a **high mobility group (HMG) box** — a conserved DNA-binding motif that **kinks DNA** → altered gene expression → testis formation → male phenotype.
- SRY does not act alone. **DAX1** (dosage-sensitive sex reversal, adrenal hypoplasia critical region, on chromosome X, gene 1), a nuclear hormone receptor gene, is **needed for proper testis development**, yet depending on timing can **suppress SRY** by inhibiting its downstream testis genes.
- **WNT4** (wingless-type integration site family member 4), largely confined to the adult ovary, also behaves as an **"antitestis" gene**.
- The female pathway is **no longer regarded as the passive SRY-negative "default"**: WNT4 and DAX1 can **proactively induce female gonadal development even in the presence of SRY**.

FETAL TESTIS PRODUCTS AND THEIR EFFECTS

Cell	Product	Effect
Leydig	Testosterone	Internal genitalia (Wolffian duct development)
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testis migration towards the scrotum
Target tissue (5 α -reductase)	DHT	Masculinises the genital anlage → external genitalia
Sertoli	Müllerian-inhibiting substance (MIS; anti-Müllerian hormone, AMH)	Prevents Müllerian duct forming uterus and fallopian tubes; keeps early germ cells quiescent (inhibits meiosis)

- Deficiencies in these pathways produce **birth defects or intersex disorders**.
- Feedback relationships are **established during gestation**; **kisspeptin** helps activate GnRH neurones and trigger GnRH release; **steroidogenic factor 1 (SF-1)**, an orphan nuclear receptor from developing Sertoli cells, contributes to axis development.

Postnatal sequence: placental steroid withdrawal at birth → **neonatal high gonadotrophin secretion** → axis sensitivity rises → FSH and LH fall to **low childhood levels** → **puberty begins with GnRH pulsing** (hypothalamic capacity usually from about the **12th year**) → gonadotrophins reach adult levels → sex hormones rise.

- Puberty begins at **critical growth, weight and nutritional rates**; likely initiated by **kisspeptin, melatonin and leptin**. **Leptin** (adipocyte hormone) signals fat-store size and modulates hypothalamic and pituitary activity.

Concept — *male development is built hormone by hormone — testosterone the ducts, DHT the external genitalia, MIS the removal of Müllerian structures — and sex determination itself is a contest between pro-testis SRY and active anti-testis genes, not a default.*

Aging and the Hypothalamic-Pituitary-Gonadal Axis

- **Progressive decline in testosterone and sperm production:** mean plasma testosterone in the seventh decade is about one-third lower than in young men.
- Terms: male menopause, male climacteric, andropause; more appropriately **partial androgen deficiency in the aging male (PADAM)**.
- Seminiferous epithelium: **decreased tubule volume and length**; lower sperm output from **decreased germ cell proliferation, not increased degeneration**; FSH rises (about threefold).

- Causes are multifactorial: **fewer Leydig cells** and **more testosterone-binding proteins**; **loss of diurnal variation**; **blunted feedback** to low testosterone (despite generally high gonadotrophins) and to GnRH; pulsatile GnRH replaced by **irregular, less effective pulses**.

Concept — ageing is a double failure — *peripheral* (fewer Leydig cells, reduced proliferation, rising FSH) and *central* (blunted feedback, disordered pulsatility) — so neither compartment alone explains the decline.

KEY POINTS: HYPOTHALAMIC-PITUITARY-GONADAL AXIS

- Normal testosterone and sperm production depend on **pulsatile secretion** of hypothalamic GnRH and pituitary LH and FSH.
- Regulation of HPG hormones occurs primarily through **negative feedback**.
- Maleness derives from **SRY** on the Y chromosome; **WNT4** and **DAX1** are antitestis genes that can proactively induce female gonadal development.
- With paternal age: **lower testosterone, blunted feedback, irregular pulsatility**.

TESTIS

Gross Architecture

- White, ovoid; **15–25 mL** volume; **4.5–5.1 cm** long.
- **Tunica albuginea** contains **smooth muscle** within collagen: may give the capsule contractility, affect testicular blood flow and promote outflow of seminiferous tubule fluid.
- Septa divide the parenchyma into **lobes, each with a centrifugal artery**. Tubules harbour developing germ cells.
- **Interstitial tissue**: Leydig cells, mast cells, macrophages, nerves, blood and lymph vessels — **20–30% of testicular volume**.
- Seminiferous tubules are long, coiled and looped, **both ends terminating in the rete testis**; **600–1200 tubules**, combined length about **250 m**.
- The **rete testis** (the "hub") coalesces into **6–12 ductuli efferentes** → caput epididymis.

ARTERIAL SUPPLY TO TESTIS AND EPIDIDYMIS

Artery	Origin / relation
Internal spermatic (testicular)	Abdominal aorta ; intimately associated with the pampiniform plexus
Deferential (vasal)	Accompanies the vas; richest collateral link to the testicular artery
External spermatic (cremasteric)	Collateral source

- **Countercurrent exchange** in the pampiniform plexus (counterflowing artery and veins) exchanges heat and small molecules — e.g. testosterone diffuses passively from vein to artery (concentration-limited).
- Arterial blood reaching the testis is **2–4 °C below rectal temperature**; **loss of this differential** is associated with testicular dysfunction in **varicocele** and **cryptorchidism**.
- **Varicocele surgery relevance**: arteries in the spermatic cord — **single in about half of cords, two in about 30%, three in about 20%**.
- Below the pampiniform plexus near the mediastinum the artery is highly coiled and branches before entry; at the testis a **single artery most often enters**, otherwise two or more branches (angiography).
- **Collaterals**, especially internal spermatic–deferential, maintain viability **after division of the internal spermatic artery**; but where there is a **single testicular artery, its interruption can cause atrophy**.

Intratesticular course: arteries pierce the tunica albuginea → run **inferiorly along the posterior surface** within the parenchyma → branches pass **anteriorly** over the parenchyma; major branches also cross the **inferior pole** and fan over the anterior surface. These vessels are at risk at **orchidopexy, testis biopsy and sperm retrieval**; the **midsection has relatively fewer vessels** than superior or inferior areas.

- **Centrifugal arteries** run in the septa → arterioles → two capillary beds, stated as a pair: **intertubular** (in the columns of interstitial tissue) versus **peritubular** (ladder-like, alongside the tubules).
- Testicular blood flow about **9 mL/100 g/min**.
- **Intratesticular veins do not run with the arteries:** parenchymal veins drain to surface veins or to a mediastinal group along the rete; these join deferential veins to form the **pampiniform plexus**, whose **thin walls** favour diffusion of testosterone and heat to the gonadal artery.
- **No known somatic innervation;** autonomic supply from the **intermesenteric nerves and renal plexus** along the testicular artery. Adrenergic fibres are restricted mainly to small vessels supplying **Leydig cell clusters** and may regulate steroidogenesis.
- Vascular tone is regulated at several levels — capsular artery autoregulation, regional metabolic control by peptides such as **atrial natriuretic peptide**, and assisted transendothelial transport of LH — a highly specialised microvasculature.
- Prominent **lymphatics** in the cord; their obstruction dilates the **interstitium but not the tubules** (the interstitium is lymph-drained) and causes **hydrocele** — **a known complication of varicocele and herniorrhaphy**.
- **Intratubular fluid** bathes Sertoli cells, flows tubule → rete testis → caput; **isosmotic with plasma**, mainly of tubular origin; its **reabsorption in the rete and efferent ductules is regulated by oestrogens**. Its composition differs markedly from plasma and lymph → basis of the **blood-testis barrier**.

Concept — *the testis is engineered to run cool and sealed: countercurrent pampiniform exchange keeps it 2–4 °C below core and the tubules keep their own fluid; varicocele, cryptorchidism, post-varicocele hydrocele and atrophy after single-artery ligation are each failures of that engineering.*

Testis Cytoarchitecture

Interstitium

Leydig Cells

- Interstitium contains blood vessels, lymphatics, fibroblasts, macrophages, mast cells and **Leydig cells**, which produce the **bulk of testicular steroid**.
- Leydig cells differentiate from **mesenchymal precursors by the seventh week of gestation**; activation of steroidogenesis coincides with androgen-dependent differentiation of the reproductive tract.
- Differentiation is driven by **LH and placental human chorionic gonadotrophin (hCG)** and local paracrine factors such as **insulin-like growth factor-1 (IGF-1)**.
- **By 3 months after birth**, a **second wave** of Leydig differentiation in response to pituitary LH briefly elevates testosterone; neonatal androgen **imprints the hypothalamus, liver and prostate** for appropriate later androgen response.
- After pubertal reactivation, a young adult testis contains about **700 million Leydig cells**.

Testosterone

- Testosterone, synthesised from cholesterol, is the **principal testicular steroid**; numerous C18, C19 and C21 steroids are also produced.

Synthesis: cholesterol stores (LH mobilises them via **cholesterol esterase**) → transport to the inner mitochondrial membrane → **side-chain cleavage enzyme** → **pregnenolone** → out of the mitochondrion into

smooth endoplasmic reticulum → **testosterone** → diffuses out of the cell → held in extracellular fluid and plasma by **steroid-binding proteins**.

MITOCHONDRIAL CHOLESTEROL TRANSPORT PROTEINS

Steroidogenic acute regulatory (StAR) protein	Peripheral benzodiazepine receptor (PBR)
Synthesis induced by LH binding ; threads through the outer mitochondrial membrane to facilitate cholesterol transport	Forms a channel for cholesterol in the mitochondrial membrane; functional interaction with StAR unclear

MAJOR ENZYMES OF TESTOSTERONE BIOSYNTHESIS

Enzyme	Step
Cholesterol side-chain cleavage enzyme	Cholesterol → pregnenolone (mitochondrion)
3 β -Hydroxysteroid dehydrogenase	Pregnenolone → testosterone (smooth endoplasmic reticulum)
Cytochrome P450 17 α -hydroxylase/C17-20-lyase	
17 β -Hydroxysteroid dehydrogenase	

- Mutations in these genes cause **disorders of androgen biosynthesis** — a **relatively rare cause of sexual ambiguity** in chromosomally normal males.

Control of Testosterone Synthesis

REGULATORS OF LEYDIG CELL STEROIDOGENESIS

Category	Factors
Principal regulator	LH → cAMP → cholesterol transport into mitochondria
Pituitary modifiers of the LH response	FSH, prolactin
Non-pituitary modifiers	GnRH; inhibin and activin; epidermal growth factor (EGF), IGF-1, TGF- β ; prostaglandins; adrenergic stimulation
Direct inhibitors	Oestrogens and androgens

Testosterone Cycles

TESTOSTERONE PEAKS AND THEIR DEVELOPMENTAL ROLE

Peak	Timing	Developmental event
Fetal	12–18 weeks' gestation	Development of the fetal reproductive tract
Neonatal	About 2 months of age	"Imprinting" of androgen-dependent target tissues
Pubertal	Second decade (rise at 12–17 years)	Masculinisation
Adult	Maximum in second–third decade → plateau → slow decline; dramatic fall in senescence	Maintenance of androgen-dependent organs

- Superimposed rhythms: **annual** (peak **autumn**, nadir **spring**), **daily** (peak **morning**, nadir **evening**), and frequent **irregular fluctuations** — reflecting continuous pituitary–testis interaction.
- **Figure 67.9 legend** (Ewing, Davis and Zirkin, 1980): testosterone is low through the prepubertal period between the neonatal and pubertal peaks.

Concept — every testosterone peak has a developmental job — build, imprint, masculinise, maintain — and all are driven through one LH-stimulated, StAR-dependent step: cholesterol entry into the mitochondrion.

Seminiferous Tubules

- A unique environment for gamete production: **support cells** (Sertoli cells; fibrocytes and myoid cells of the basement membrane) and **germ cells** (slowly dividing stem cells → rapidly proliferating spermatogonia and spermatocytes → metamorphosing spermatids).

Sertoli Cells

- Line the tubules, resting on the basement membrane with cytoplasmic ramifications to the lumen; irregular nuclei, prominent nucleoli, **low mitotic index**.
- **Tight junctions between adjacent Sertoli cells** — the **strongest intercellular barriers in the body** — divide the tubule into **basal and adluminal compartments**, the basis of the **blood–testis barrier** and of an **immunologically privileged site**.
- Sertoli cells are **nurse cells**: germ cells are nourished within and between their cytoplasmic projections; a **polarised epithelium** whose base faces the plasma environment and whose apex holds the unique tubular environment.

BASAL VERSUS ADLUMINAL COMPARTMENT

Basal	Adluminal
Beside basement membrane; plasma-like environment	Towards lumen; environment unique to the tubule
Spermatogonia and early (preleptotene) spermatocytes	All later germ cells — spermatocytes from leptotene–zygotene onwards, spermatids

- Nurturing functions: **specialised adluminal microenvironment**; **gap junctions** with germ cells; **migration** of germ cells — tight junctions are **constantly remodelled**, opening and closing to let germ cells pass. Ligand–receptor systems such as **c-kit and kit ligand** mediate Sertoli–germ cell communication.
- Sertoli cells also **phagocytose germ cells** and secrete fluid and effector molecules.
- **ABP**, one of the earliest described Sertoli products, is an intracellular androgen carrier that keeps **intratubular androgen about 50-fold above serum**; testosterone in turn regulates Sertoli function, including ABP production.
- **Inhibin B**: Sertoli-derived, negative feedback on FSH; an **endocrine marker of Sertoli cell function** in male infertility evaluation.
- Further secretions, beyond the FSH-induced products above: extracellular matrix (**laminin, collagen types IV and I**); glycoprotein 2, somatomedin-like substances, T proteins, H-Y antigen, cyclic proteins; and steroids — DHT, testosterone, androstenediols, 17 β -oestradiol, other C21 steroids (roles largely unknown).

Germ Cells

- Output about **123 million spermatozoa daily** (wide range) — about **1000 per heartbeat**; germ cells are **ordered from basement membrane to lumen** in at least **13 recognisable types**, each a step in spermatogenesis.

Sequence (least → most differentiated): dark type A spermatogonia (Ad) → pale type A (Ap) → type B (B) → primary spermatocytes: preleptotene (R) → leptotene (L) → zygotene (Z) → pachytene (P) → secondary spermatocytes (II) → spermatids Sa → Sb → Sc → Sd₁ → Sd₂.

- **Figure 67.11 legend** (after Clermont, 1966): six stages (I–VI) of the cycle; **Ad and Ap spermatogonia appear in every stage**; diakinesis (D₁) in stage V; both maturation divisions (I_D, II_D) in stage VI; residual body (Rb) shed with Sd₂.
- **Figure 67.10 legend** (Russell, 1980): Sertoli cell as a tree with trunk and limbs; basal, intermediate and adluminal compartments. Junctions form **above and below leptotene–zygotene spermatocytes** as they translocate (intermediate compartment) → spermatocytes enter the adluminal compartment when the upper junctions dissociate → elongating spermatid sits in a recess of the trunk → advanced spermatid

moves luminally for spermiation, only its head in contact. Contacts: desmosome–gap junction complexes, ectoplasmic specialisations, tubulobulbar complexes.

Peritubular Structure

LAYERS OF THE PERITUBULAR TISSUE

Layer	Content
Outer adventitial	Fibrocytes (fibroblasts)
Middle	Myoid cells with connective tissue lamellae; abundant microfibrils
Inner	Collagen matrix (inner lamella)

- **Myoid cells** are thought to be **contractile**; they secrete extracellular matrix (**fibronectin, collagen type I**) and produce the inner collagenous layer.
- Myoid cells interact precisely with Sertoli cells (mesenchymal–epithelial): the myoid paracrine factor **P-Mod-S (peritubular modifies Sertoli)** profoundly affects Sertoli synthetic function in vitro.

Blood-Testis Barrier

- Injected dyes reach all tissues except regions of **brain and testis**. More accurately termed the **blood–seminiferous tubule barrier**; it has two components.

ANATOMICAL VERSUS FUNCTIONAL BARRIER

Anatomical (mechanical)	Functional (immunological)
Partly from myoid cells ; traffic also regulated at capillary endothelium	Restricted lymphocyte types and numbers in vulnerable regions of the germinal epithelium
Most important: Sertoli–Sertoli tight junctions , excluding large molecules and lymphocytes	Immunological tolerance : small continuous leak of sperm antigen in weaker areas (rete testis, efferent ductules, epididymis) → T-suppressor cells and tolerance, like allergen desensitisation; a large antigen load provokes a true immune response
Necessary but not sufficient — absent in other protected areas of the tract	Sertoli immunosuppressive mediators: galectin-1 ("master immune regulator"), interferon- γ , soluble Fc receptor, TGF- β , androgens

- The barrier develops at **spermarche**; likely important for **meiosis** and for protecting gametes that are **not recognised as self**.
- Its value is realised **only after puberty**, when postmeiotic "foreign" antigens first exist: testicular insult (biopsy, torsion, trauma) **before puberty does not induce antisperm antibodies**; **after puberty, immunological infertility is a known risk**.
- Clinically, it may **limit chemotherapy access** to intratesticular cancer cells → **isolated testicular recurrence**.

Concept — *the barrier exists because postmeiotic germ cells appear only after the immune system has learned self — so it matters only after puberty, and it is active tolerance as much as a physical seal.*

Spermatogenesis

- A complex process of **DNA reduction and germ cell metamorphosis**. Classically about **64 days**; an in vivo labelling study found the time to produce an ejaculated sperm ranges **42–76 days** — wide interindividual variation.
- **Figure 67.13 legend**: deuterated water labelling in men with normal semen quality; new ejaculated sperm appeared **as early as 42 days**.

Three phases: **proliferative** (spermatogonia self-renew or differentiate) → **meiotic** (reduction division → haploid spermatids) → **spermiogenesis** (metamorphosis of spermatids into spermatozoa).

CYCLE, STAGE AND WAVE

Term	Meaning
Cycle	Division of primitive spermatogonial stem cells into subsequent germ cells; several cycles coexist
Stage	Cellular association seen from one fixed point in a tubule — six in humans
Wave	Spatial organisation of cycles along the tubule — helical/spiral in humans , making production continuous, not pulsatile

Testis Stem Cell Migration, Renewal, and Proliferation

Testis Stem Cell Migration

Primordial germ cells migrate to the **gonadal ridge** → associate with Sertoli cells in primitive cords → called **gonocytes** once the gonad forms seminiferous cords → called **spermatogonia** after migrating to the tubule periphery.

- Early migrating germ cells resemble **embryonic stem cells** and are likely the **source of adult germ cell tumours**.
- **Failure to migrate** into the primitive testis is thought to cause **extragonadal germ cell tumours** and **azoospermia with Sertoli cell-only histology**.

Testis Stem Cell Renewal

- Spermatogonia in the **stem cell niche** are replenished (**stem cell renewal**) via the **kit ligand/c-kit** system and the niche factor **glial cell line-derived neurotrophic factor (GDNF)**.
- Human spermatogonial stem cells from adult biopsies can be **reprogrammed in vitro to embryonic-like stem cells**: express pluripotency markers (octamer-binding transcription factor 4 [OCT-4], sex-determining region Y-box 2 [SOX-2], STELLAR, growth differentiation factor 3 [GDF-3]), form all three germ layers, keep a normal karyotype, form teratomas, and express epigenetic markers and telomerase → possible **patient-specific cell therapy**.
- **Figure 67.15 legend**: colonies grown under human embryonic stem cell conditions show the **cobblestone appearance** of embryonic stem cells and are functionally multipotent.

Testis Stem Cell Proliferation

- **Ap spermatogonia** in the basal niche **divide every 16 days** to form **B spermatogonia**, which are **committed** to become spermatocytes.
- Daughter cells remain joined by **cytoplasmic bridges**, seen between germ cells of **all classes** — probably for **synchronised proliferation, differentiation and gene regulation**.

Testis Stem Cell Aging or Senescence

- Renewal continues from **puberty to the eighth decade** — reproductive longevity at a cost:
- (1) **Paternal age effect (PAE) disorders** — gamete quality control declines, more disease in offspring.
- (2) **"Selfish spermatogonia"** — non-random mutations in the **fibroblast growth factor receptor 2 and 3 (FGFR2, FGFR3) RAS-receptor pathways** give clonal advantage; one consequence is oncogenesis — the basis of **spermatocytic seminoma in men of advanced reproductive age**.

Meiosis

- **Somatic cells replicate by mitosis** (DNA duplication → nuclear envelope breakdown → equal division; identical daughters); **germ cells by meiosis**, which generates **genetic diversity** for natural selection.
- **Essential difference**: after a single DNA duplication, meiosis divides **twice**, so a diploid (2n) parent yields haploid (n) gametes.

TABLE 67.2 ESSENTIAL DIFFERENCES: MITOSIS AND MEIOSIS

	Mitosis	Meiosis
Cells	Somatic	Sexual (germ)
Divisions / daughters	One / two	Two / four
Chromosome number	Maintained	Halved
Homologues	No pairing	Synapsis in prophase I
Crossovers	None	More than one per homologue pair
Centromere division	Anaphase	Anaphase II
Daughter genotype	Identical	Genetically varied

- The table isolates the examinable asymmetry: **synapsis and crossing over occur only in meiosis**, and in meiosis centromeres do not divide until the **second** division.
- **Small RNAs** — small interfering RNAs (siRNAs), microRNAs (miRNAs), piwi-interacting RNAs (piRNAs) — regulate germ cell gene expression at the **post-transcriptional or translational** level.

Sequence: type B spermatogonia divide **mitotically** → **primary spermatocytes** (the first germ cells to undergo meiosis; they cross from the basal to the adluminal compartment) → **first meiotic division — the reduction division** → secondary spermatocytes → **second, mitosis-like (equational) division** → **haploid Sa spermatids**, each genetically distinct because of recombination.

- **Chromosomal recombination** is the **defining feature of mammalian meiosis** and the engine of genetic diversity: in meiotic prophase a **synaptonemal complex** pairs maternal and paternal homologues, which exchange DNA at crossing-over sites (**chiasmata**).
- **Defective recombination causes azoospermia**: significant defects are found in a subset of non-obstructive azoospermic men and in **about half of men with maturation arrest** — faulty recombination is linked to poor sperm production.
- Altered crossover position is a risk factor for **nondisjunction** → **sperm aneuploidy**; the association in azoospermic men is strong enough to explain the **higher rate of chromosomal abnormality in offspring from in vitro fertilisation (IVF) and intracytoplasmic sperm injection (ICSI)**.

Spermiogenesis

- Round **Sa spermatids become spermatozoa without dividing**.

CHANGES DURING SPERMIOGENESIS

Structure	Change
Cytoplasm	Lost; organelles migrate
Acrosome	Formed from the Golgi apparatus
Flagellum	Formed from the centriole
Nucleus	Compacted to about 10% of former size ; spherical → asymmetrical as chromatin condenses
Mitochondria	Reorganised around the midpiece

- Nuclear reshaping involves chromosome structure and proteins, the perinuclear cytoskeletal theca, the **manchette** of microtubules, subacrosomal actin and Sertoli interactions.
- After elongation the **Sertoli cytoplasm retracts**, stripping residual cytoplasm and **extruding the sperm into the lumen**. Production reaches **up to 300 sperm per gram of testis per second**.

Sertoli Cell–Germ Cell Interaction

- Cell–cell networks: Leydig–Sertoli, Leydig–peritubular, Sertoli–peritubular and Sertoli–germ cell.

- **TGF- β 3 and tumour necrosis factor- α (TNF- α)** can reversibly disrupt the blood–testis barrier by reducing **occludin and zonula occludens-1 (ZO-1)** via a **p38 mitogen-activated protein (MAP) kinase** pathway.
- **Figure 67.14 legend** (Schulze and Rehder, 1984): **helical configuration** of epithelial cycles forms overlapping waves that keep sperm production constant.

Concept — *spermatogenesis is continuous because it is staggered in space — cycles arranged as helical waves — while each clone is synchronised internally by cytoplasmic bridges; its weak points are recombination in meiosis and the lifelong mitotic self-renewal that accumulates mutations.*

Genetics

- Genetic causes of abnormal spermatogenesis: **single-gene mendelian mutations** (e.g. cystic fibrosis) and **chromosomal disorders** (structural or numerical).
- Deletions of the Y long arm were postulated decades ago to cause azoospermia: the **azoospermia factor (AZF)** region, now subdivided by microdeletion pattern into **AZF_a, AZF_b and AZF_c**.
- **Yq microdeletions** occur in **6–8% of severely oligospermic** and **15% of azoospermic** men — the **most commonly defined molecular cause of male infertility**.

AZOOSPERMIA FACTOR DELETIONS: HISTOLOGY AND SPERM RECOVERY

Deletion	Typical histology	Sperm in ejaculate or testis
Complete AZF_a	Germ cell aplasia / Sertoli cell-only	Highly unlikely
Complete AZF_b	Maturation arrest — primary spermatocyte (early) or spermatid (late)	Highly unlikely ; isolated ejaculated cases
AZF_c (partial or complete)	Hypospermatogenesis , or Sertoli cell-only with foci of spermatogenesis	Often found
Partial AZF _a or AZF _b	—	Ejaculated sperm detected
AZF _a +b, AZF _b +c (presumed partial)	—	Ejaculated sperm reported
Complete AZF_a–c	—	Azoospermia; none on biopsy

- **Pattern:** prognosis worsens proximally — AZF_c is the retrievable deletion, complete AZF_a and AZF_b are not.
- The **X chromosome** also matters: many genes expressed exclusively in mouse spermatogonia map to X; candidate human X-linked fertility genes include **SOX3** (sex-determining region Y-box 3), **TEX11** (testis-expressed 11) and **FATE** (fetal and adult testis-expressed) — X genes may explain a share of unexplained infertility.

Genetics and Paternal Age

Age-Related Sperm Chromosomal Anomalies

- Studied first because of concern about **trisomy 21**. **Fluorescence in situ hybridisation (FISH)** shows the PAE **increases sex-chromosome aneuploid sperm**, but there is **little evidence of more aneuploid births**, except possibly trisomy 21 and disomy 1 (very rare).
- **Structural anomalies in sperm rise linearly with paternal age** — continued division places germ cells at risk; yet, **except for reciprocal translocations**, little evidence of more de novo structural anomalies in offspring.

Age-Related Sperm Genetic Mutations

- Single-gene defects arise from **DNA replication errors**. The disorders in Box 67.1 are **sentinel phenotypes**: significant frequency, **low fitness**, **highly penetrant** mutations.

- **Mechanism:** about **30 spermatogonial divisions by puberty**, then **23 divisions per year** — the more divisions, the more replication errors.
- **Risk of a new autosomal dominant mutation:** **0.22 per 1000 births under 29 years** → **doubles (0.45) at 40–44** → **3.7 per 1000 over 45**.

BOX 67.1 GENETIC DISORDERS IN OFFSPRING ASSOCIATED WITH ADVANCED PATERNAL AGE (GROUPED)

Group	Disorders
Skeletal and craniofacial	Achondroplasias, Apert syndrome, Crouzon syndrome, Treacher Collins syndrome, oculodentodigital syndrome, fibrodysplasia ossificans, Marfan syndrome
Tumour predisposition	Bilateral retinoblastoma, neurofibromatosis, tuberous sclerosis, polyposis coli
Other single-gene	Aniridia, Waardenburg syndrome, progeria, polycystic kidney disease; X-linked: haemophilia A, Lesch-Nyhan syndrome
Neurodevelopmental / psychiatric	Schizophrenia, bipolar disorder, autism, attention deficit disorders , Alzheimer disease (postulated)

- **Pattern and trap:** the list is dominated by **de novo dominant** conditions (FGFR-pathway skeletal dysplasias above all) but also contains X-linked and complex neuropsychiatric disorders — the latter link to the epigenetic data below.

Age-Related Sperm Epigenetic Changes

- **Sperm DNA methylation marks change with age:** hundreds of defects accumulate, **non-randomly**, at sites controlling **neurodevelopmental genes**.
- In men donating sperm when young and again when older, the **epigenetic change rate in sperm was about double** that estimated for other tissues; changes **cluster in schizophrenia and bipolar genes** → at least some PAE disorders may be **epigenetically transmitted**.

Age-Related Sperm DNA Fragmentation Changes

- **Sperm DNA fragmentation rises with paternal age** — single- and double-strand breaks; conceptually like programmed cell death but **induced by oxidative stress after sperm release from Sertoli cells**.
- Associated with **impaired fertility, poor early embryo development and miscarriage**, probably because the oocyte cannot correct the volume of breaks introduced into the embryonic genome.

Concept — *the male germline never stops dividing, and every paternal-age effect — point mutation, selfish clones, aneuploidy, methylation drift, fragmentation — is the price of lifelong spermatogonial self-renewal.*

KEY POINTS: TESTIS

- The testis contains **250 m of seminiferous tubules** and **700 million Leydig cells** in the young adult.
- Spermatogenesis occurs in **stages, cycles and waves** to ensure constant sperm production.
- With paternal age: more sperm **structural chromosomal abnormalities, autosomal dominant mutations and epigenetic alterations**.
- **Sperm DNA fragmentation** increases with paternal age and contributes to decreased fertility.
- Strong association between advanced paternal age and **neurodevelopmental sentinel phenotypes** in offspring.

EPIDIDYMIS

Gross Architecture

- **Comma-shaped** organ on the **posterolateral** surface of the testis.

- Passage converts **testicular sperm, which have very poor or no motility**, into sperm with functional motility, **chemotaxis to follicular fluid** and fertilising ability, through membrane, charge and metabolic changes (see Sperm Biochemical Changes).
- A **3–4 m duct**, tightly coiled within a **tunica vaginalis sheath**; septa from the sheath divide it into **caput, corpus and cauda**.
- **Caput: 8–12 ductuli efferentes** — lumen large and irregular near the testis, narrowing and oval near the ductus epididymidis → coalesce into a **single duct in the corpus** → in the bulky **cauda** the tubule enlarges, becomes irregular and gradually assumes the appearance of the **vas deferens**.

Vascular and Lymph Supply

- **Caput and corpus**: branch of the **testicular artery** → superior and inferior epididymal branches; also deferential branches, with collaterals to the testicular supply.
- **Cauda**: branches of the **deferential artery**.
- The **deferential and cremasteric arteries are collateral sources** when the main testicular artery is obstructed or ligated.
- Arteries enter along septa, **coil extensively**, then become straight microvascular vessels. **Capillary density is greatest in the proximal caput**, less distally; the network is under steroid regulation (animal data).
- **Venous**: corpus and cauda veins form the **vena marginalis epididymis of Haberer** → pampiniform plexus via the vena marginalis testis, or via cremasteric or deferential veins.

LYMPHATIC DRAINAGE OF THE EPIDIDYMIS

Caput and corpus	Cauda
As for the testis — beside the internal spermatic vein → preaortic nodes	With the vas deferens lymphatics → external iliac nodes

Innervation

- **Intermediate spermatic nerves** (superior hypogastric plexus) and **inferior spermatic nerves** (pelvic plexus).
- Ductuli efferentes and proximal epididymis: **sparse sympathetic** fibres, mainly perivascular. Nerve density **increases progressively distally**, together with the appearance of smooth muscle.
- This explains **rhythmic peristalsis** of the ductuli and **intermittent contraction of the cauda and vas during emission** — critical for sperm movement.

FIGURE 67.16 LEGEND — EPIDIDYMAL SEGMENTS (BAUMGARTEN, HOLSTEIN AND ROSENGREN, 1971)

Segment numbers	Region
1a–d	Ductuli efferentes
2a–b	Ductus epididymidis — caput
3a–c	Ductus epididymidis — corpus
4a–c	Ductus epididymidis — cauda
5 / 6	Ductus deferens, pars epididymica / pars libera

Cytoarchitecture

Epididymal Epithelium

PRINCIPAL VERSUS BASAL CELLS

Principal cells	Basal cells
Majority	Far fewer

Principal cells	Basal cells
Height varies with stereocilia (microvilli, not cilia) — tall proximally (about 120 µm), shorter distally (about 50 µm)	Tear-shaped , on the basal lamina; apices form threads between principal cells
Absorptive and secretory : coated pits, micropinocytotic and multivesicular bodies, membranous vesicles, extensive Golgi — capacity varies along the duct	Thought to derive from macrophages ; possible precursors of principal cells
Morphology varies along the duct	Morphology constant along the duct

- Regional epithelium: the **epididymis proper** is **pseudostratified** (principal and basal cells); the **ductuli efferentes** have **ciliated cells** (conduct sperm into the epididymis) and **non-ciliated cells** (secretory; some with microvilli, resorptive).
- Apical junctional complexes suggest a **blood–epididymis barrier**, less tight than the blood–testis barrier, extending **caput to cauda** and shaping the composition of luminal fluid in each segment.
- **Figure 67.17 legend**: electron micrograph — principal cells, basal cells, stereocilia and myofilaments.

Epididymal Contractile Tissue

CONTRACTILE LAYER ALONG THE DUCT

Region	Contractile tissue
Ductuli efferentes	Loose layer 2–4 cells deep ; myofilaments, nexus-like junctions
Corpus	Larger contractile cells resembling smooth muscle
Cauda	Thick smooth muscle in three layers — longitudinal, circular, longitudinal — thickening as it becomes the vas

Concept — *along the duct, muscle and sympathetic innervation increase together from caput to cauda, so transport grows more active and storage more controllable distally.*

Epididymal Function

- Variation in anatomy and histology suggests **several functional tissues**; epididymal passage produces sperm transport and storage, fertilising ability, chemotaxis and motility maturation.

Sperm Transport

- Transit takes **2–12 days**; caput–corpus transit ≈ cauda transit; related to **daily testicular sperm production**, not age or ejaculation frequency.
- Sexual frequency **does not alter caput and corpus transit**, but **recent emissions markedly shorten cauda transit**.

Mechanism (sperm are immotile on entry and remain so in the caput): **rete testis fluid** carries sperm into the ductuli → flow drawn on by **fluid resorption** by ductal epithelium, mediated by the **oestrogen receptor** → **motile cilia and myoid contraction** in the ductuli → within the epididymis proper, **spontaneous rhythmic contraction of the duct's contractile cells** — the principal mechanism.

Sperm Storage

- Sperm are held in the **cauda** for variable periods depending on sexual activity; in men under 55, each epididymis holds **155–209 million sperm, about half in the cauda**.
- **Cauda sperm**, unlike testicular sperm, show **progressive motility, chemotaxis to follicular fluid and fertilising ability**.
- Animal sperm stay viable **several weeks after vasal ligation**, but fertility **declines with prolonged storage**; in humans, extended transit and storage may raise **DNA fragmentation** and reduce fertility.

- Fate of unejaculated sperm is uncertain: in animals — spontaneous discharge, oral self-cleaning, urine, epididymal reabsorption; in humans, **spermiphages** (luminal macrophages) phagocytose sperm after vasal ligation; their role in unvasectomised men is unclear.

Sperm Maturation

Sperm Motility

- Motility capacity is **gained during epididymal migration**.

MOTILITY PATTERNS BY REGION (FIGURE 67.18)

Region	Pattern
Ductuli efferentes, initial segment	Immature: immotile or weak twitching; high-amplitude, low-frequency beats, little forward progression
Corpus	Rising fraction mature
Cauda	Majority mature: low-amplitude, high-frequency beats → progressive motility

- Contact time versus region is unresolved: animal data suggest partly **intrinsic** maturation; in obstruction (vasal or epididymal, congenital absence of the vas) sperm from the **distal epididymis move poorly and proximal sperm better** → motility develops with **contact time with proximal epithelium**, though obstructed maturation may differ from normal.

Sperm Fertility

- **Testicular sperm cannot fertilise** unless injected by micromanipulation.
- Fertilising ability is **acquired gradually**: rabbit sperm show a steep rise from caput to corpus to cauda; zona-free hamster egg studies corroborate. **Proximal human sperm can bind zona-free eggs; only cauda sperm penetrate**.
- **Fertility maturation is achieved mostly at the late corpus or early cauda** (Figure 67.19: fertilising ability rises from near zero in the caput to maximal in the cauda).
- Challenge: **natural pregnancies follow vasoepididymostomy at the level of the ductuli efferentes** — obstruction may skew maturation proximally, or slower flow may allow more contact time.
- Nonetheless, **fertility is more likely the more distal the anastomosis**.
- After reversal of **vasectomies over 15 years old**, ejaculated **concentration is maintained but motility falls** — acquired epididymal dysfunction from prolonged obstruction affects reversal outcome.

Sperm Biochemical Changes

- Transit confers a **net negative surface charge**; **sulfhydryl groups oxidise to disulfide bonds** → structural rigidity needed for progressive motility and egg penetration.
- Membrane changes in lectin binding, phospholipid and lipid content, glycoprotein composition, immunoreactivity and iodination → **enhanced adherence to the zona pellucida**.
- Metabolic changes: **increased glycolytic capacity**, altered intracellular pH and calcium, modified adenylate cyclase, altered phospholipid fatty acids.

Regulation of Epididymal Function

- Luminal fluid differs from serum and **varies by region** (osmolarity, electrolytes, protein), from vascularisation, barrier activity and selective handling of **glycerylphosphorylcholine (GPC)**, **carnitine** and **sialic acids**.
- Fluid proteins active on sperm: forward motility protein, sperm survival factor, progressive motility-sustaining factor, sperm motility-inhibiting factor, acidic epididymal glycoprotein, and **EP2-EP3 proteins** (zona binding).

- **Epididymosomes** traffic the sperm **small RNA payload** that sperm deliver to the embryo and that helps drive embryo development → the epididymis is an important potential source of sperm dysfunction.
- **Androgen dependence:** testosterone and DHT are very concentrated in the epididymis; the **castrated epididymis loses motility support, fertility maturation and storage**, all reversed by androgen.
- The epididymis needs **higher androgen levels than other accessory glands**; action is mainly via **DHT** (primary tissue androgen) and/or 5 α -androstane-3 α ,17 β -diol (3 α -diol); Δ^4 -5 α -reductase and 3 α -hydroxysteroid dehydrogenase are present → explains **impaired semen quality with 5 α -reductase inhibitors**.
- **Temperature:** chronic heat (intra-abdominal placement) abolishes storage and electrolyte transport → a mechanism for **varicocele and cryptorchidism**.
- **Innervation:** partial denervation in the rat → sperm accumulate in the cauda and swim more slowly → relevance to neuropathic infertility (**spinal cord injury, diabetes mellitus**).

Concept — the epididymis turns immotile, infertile testicular sperm into motile, fertile gametes by time and place — fertility arrives at the late corpus or early cauda, which is why distal anastomoses fare better — and it depends on DHT, a cool scrotum and intact nerves.

KEY POINTS: EPIDIDYMIS

- **Principal cells** (absorptive and secretory), **basal cells** (macrophage-derived) and **contractile cells** (sperm transport).
- Sperm mature by gaining **progressive motility, chemotaxis** and the ability to **bind and penetrate the zona pellucida**.
- Function is **temperature and androgen dependent** — relevant to cryptorchidism, varicocele and 5 α -reductase inhibitor use.

DUCTUS (VAS) DEFERENS

Gross Architecture

- Derived from the **mesonephric (Wolffian) duct**; **30–35 cm** long, from the cauda epididymis to the **ejaculatory duct**, medial to the seminal vesicle and posterior to the prostate.
- **Five regions:** sheathless epididymal segment (within tunica vaginalis) → scrotal → inguinal → retroperitoneal (pelvic) → **ampulla**.
- Layers: outer **adventitia** (vessels, small nerves) → **muscular coat: inner longitudinal, middle circular, outer longitudinal** → mucosa with epithelium.
- Outer diameter **1.5–3 mm**; unobstructed lumen **200–700 μ m**.
- Blood supply: **deferential artery, a branch of the superior vesical artery**; venous drainage corresponds.
- Innervation: sympathetic and parasympathetic; **cholinergic supply unimportant for motor activity**. Rich **adrenergic** supply from the **hypogastric nerve via the presacral nerve**, in all three muscle layers, **densest in the outer longitudinal layer**; short adrenergic neurones; abundant **purinergic receptors** → **sympathetic–purinergic cotransmission** in transport and ejaculation.
- Neurones with neuropeptide Y, enkephalin, galanin, somatostatin, vasoactive intestinal polypeptide and nitric oxide are present; roles unclear.
- At **vasovasostomy**, testicular-side segments show **reduced noradrenergic and secretomotor nerve density** compared with abdominal segments — may affect transport and reversal success.

Cytoarchitecture

- **Pseudostratified** epithelium; **height decreases** from testis to seminal vesicle while longitudinal **folds become more complex distally**.
- Basal cells plus three tall columnar types — **principal, pencil and mitochondria-rich** — all with stereocilia and irregular convoluted nuclei; **principal cells dominate proximally**, pencil and mitochondria-rich cells **increase distally**.
- **Total muscle thickness decreases** along the length → the vas is **more than a passive conduit**.

Vas Deferens Function

Sperm Transport

- The vas shows **spontaneous motility**, responds to **stretch**, and propels fluid into the urethra by **strong peristalsis** on hypogastric nerve stimulation or adrenergic neurotransmitters → with sympathetic stimulation just before emission, sperm move rapidly from distal epididymis to ejaculatory duct.
- Consistent with the **highest muscle-to-lumen ratio (about 10:1) of any hollow viscus**.
- **Vasal sperm reserve about 130 million** — a significant share of ejaculated sperm is stored here; vasal sperm quality at vasectomy in fertile men **closely matches the ejaculate**.

VASAL TRANSPORT: SEXUAL REST VERSUS SEXUAL STIMULATION (ANIMAL DATA)

Sexual rest	After sexual stimulation
Epididymal sperm move distally and leak into the urethra in small amounts — clearing excess stored sperm	Contents propelled proximally towards the epididymis — the distal vas contracts with greater amplitude, frequency and duration
With prolonged rest, excess sperm again move distally — the vas both transports sperm and maintains epididymal reserves	

Absorption and Secretion

- **Principal cells** are typical **glycoprotein-secreting** cells; stereocilia, apical blebbing and lysosomes indicate **absorption**; **spermiphagy** by ampullary epithelium seen in men and monkeys.
- **Androgen dependent**: the vas **actively converts testosterone to DHT**; castration causes atrophy (reversed by testosterone) in monkeys and alters spontaneous and α - and β -adrenergic contractions in rats.

Concept — *once regarded as a muscular pipe, the vas is an active, androgen-dependent organ — the most muscular hollow viscus, a sperm reservoir and an absorptive-secretory epithelium.*

SEMINAL VESICLES AND EJACULATORY DUCTS

Gross Architecture and Cytoarchitecture

SEMINAL VESICLE VERSUS EJACULATORY DUCT

	Seminal vesicle	Ejaculatory duct
Form	Paired, elongated, hollow; posterior to prostate and bladder; 5–7 cm long, up to 1.5 cm wide ; one 15-cm coiled tubule	Paired, collagenous tubes from the vas-seminal vesicle junction , through the prostate, to the verumontanum
Wall	Moist folded mucosa → collagenous middle layer → circular and longitudinal muscle = 80% of wall thickness	Continuation of seminal vesicle, but the outer circular muscle does not extend into the duct ; muscle thins and dissipates distally
Epithelium	Mainly non-ciliated pseudostratified columnar or cuboidal; thin complex folds → crypts	Simple and pseudostratified columnar; complex folds

	Seminal vesicle	Ejaculatory duct
Arterial	Prostatovesical branch of the internal iliac/ inferior vesical artery (may arise from superior vesical or pudendal); anterior and posterior branches	Branches of the inferior vesical artery
Lymph	Internal iliac nodes	—
Nerves	Sympathetic: superior lumbar and hypogastric nerves; parasympathetic: pelvic plexus	Pelvic plexus

- The seminal vesicle **duct opens into the ampullary vas** as it enters the prostate.
- Ejaculatory duct regions: **proximal extraprostatic → middle intraprostatic → short distal segment** in the lateral verumontanum (Figure 67.20).
- **No valve-like muscular "sphincter"** exists at the ejaculatory duct orifice; **urinary reflux is prevented and ejaculatory continence kept by the acute angle of duct insertion** into the urethra.
- **Figure 67.21 legend** (Nguyen et al., 1996): cadaveric sections — intraprostatic ducts have a **thick collagenous layer** and only a **thin outer muscle layer**.

Seminal Vesicle and Ejaculatory Duct–Unit Function

- The seminal vesicle–ejaculatory duct unit behaves like the **bladder and urethra** (animal data): the seminal vesicle is a contractile, compliant smooth-muscle reservoir; the ejaculatory duct a urethra-like conduit.

EJACULATORY DUCT OBSTRUCTION: BLADDER ANALOGY

Physical obstruction	Functional obstruction
Blockage of the ducts — analogous to bladder outlet obstruction	Seminal vesicle dysfunction — analogous to voiding dysfunction from bladder myopathy

- **Static imaging** such as transrectal ultrasonography **may not distinguish** the two; **medications and conditions such as diabetes** may predispose to seminal vesicle dysfunction.

Seminal Vesicle Function

- Seminal vesicles contribute **about 80% of seminal fluid**, appearing in the **later fractions** of the ejaculate, after the sperm-rich epididymal and prostatic fractions.

Sperm course: ejaculation → female **cervical mucus** → uterus → **oviduct** (site of fertilisation); during residence, sperm must undergo **capacitation** — the **acrosome reaction** and **hyperactivated motility**.

Whether prostatic or seminal vesicle secretions contribute is unclear.

- The physiological role of seminal vesicle fluid is unclear; in rodents it forms a **plug** against sperm from a subsequent male.
- Semen is **liquid before ejaculation and coagulates** after mixing with seminal vesicle secretions; the major coagulum protein is **semenogelin I** (52 kDa), **expressed only in the seminal vesicles**. Coagulation may promote motility, stabilise sperm chromatin and suppress female tract immunity.
- The **best-elucidated function of semen is antioxidant protection**: enzymes **glutathione peroxidase, superoxide dismutase, catalase**; molecules **taurine, hypotaurine, tyrosine**.
- **Lipofuscin** from dead epithelial cells gives the **yellow-white colour**.
- Secretions are **alkaline** and contain **fructose** (nutrient energy for sperm), mucus, vitamin C, flavins, phosphorylcholine and prostaglandins; mixed with prostatic fluid, semen is **mildly alkaline**.
- **Acidic ejaculate (pH <7.2)** indicates **blockage or absence of the seminal vesicles**.

Concept — the seminal vesicle–ejaculatory duct unit works like bladder and urethra, so obstruction may be structural or functional; and because the vesicles supply most of the volume, the alkalinity and the fructose, their absence shows as a small, acidic ejaculate.

KEY POINTS: VAS DEFERENS, SEMINAL VESICLE AND EJACULATORY DUCTS

- The vas is of **Wolffian (mesonephric) origin** and carries sperm from the cauda epididymis to the ejaculatory duct during emission.
- The seminal vesicle–ejaculatory duct unit is **analogous to bladder and urethra** and subject to **physical blockage and functional disorders** that cause infertility.

SPERMATOZOA**Anatomy and Physiology**

- Human spermatozoon **about 60 µm** long: **head, neck, tail**.

STRUCTURE OF THE SPERMATOZOON

Part	Dimensions / limits	Contents
Head	Oval, about 4.5 × 3 µm	Nucleus with highly compacted chromatin ; acrosome — membrane-bound enzymes for penetrating the egg's outer vestments
Neck	Joins head to tail	Connecting piece and proximal centriole ; axoneme starts here
Midpiece	7–8 µm ; ends at the annulus	Axoneme + outer dense fibres + helical mitochondrial sheath (about 75 mitochondria)
Principal piece	Longest segment	Axoneme; outer dense fibre columns replaced by the fibrous sheath (longitudinal columns + transverse ribs)
Endpiece	Most distal	Axonemal structures (and fibrous sheath per source)

- **Outer dense fibres** are rich in disulfide bonds, **not contractile**; they give the tail the **elastic rigidity** needed for progressive motility.
- Except at the endpiece, a specialised **plasma membrane** regulates transmembrane movement of ions and molecules.
- **Mitochondria** produce **adenosine triphosphate (ATP)** and can trigger apoptosis by releasing **cytochrome c**; outer and inner membranes, the inner folded into **cristae** (greater surface area).
- Five inner-membrane respiratory complexes for oxidative phosphorylation: **NADH dehydrogenase** (nicotinamide adenine dinucleotide), **succinate dehydrogenase**, **cytochrome bc1**, **cytochrome c oxidase**, **ATP synthase**. Matrix: citric acid cycle, fatty acid and amino acid oxidative enzymes, ATP, mitochondrial DNA, ribosomes.
- **Mitochondrial DNA (mtDNA)**: distinct from nuclear DNA; **circular, histone-free, 16,569 base pairs**, one heavy and one light strand; encodes **respiratory complex subunits, mitochondrial rRNAs and tRNAs**; **no introns**.
- mtDNA mutates **about 40–100 times faster** than nuclear DNA: proximity to reactive oxygen species at the respiratory complexes, **no protective histones, limited repair** → hence **all paternal mtDNA is degraded in the fertilised egg**, probably via **ubiquitin-mediated proteolysis**.
- **Sperm head membrane** proteins mediate sperm–egg interaction: carbohydrate-binding proteins bind the species-specific **zona pellucida glycoprotein 3 (ZP3)** → zona binding → **acrosome reaction**.
- **PH30 (fertilin)**, present on testicular sperm and modified during epididymal transit, is the **sperm–egg membrane fusion protein**.

The Axoneme

- The **axoneme is the true motor**, requiring **200–300 proteins**; its best-understood element is the **"9 + 2" arrangement** of microtubule doublets.

- **Dynein**: a large (2000 kDa) Mg^{2+} -stimulated ATPase → ATP-driven **microtubule sliding** → axonemal bending → flagellar movement.

AXONEMAL COMPONENTS (FIGURE 67.23)

Component	Structure	Role / defect
Outer dynein arms (heavy chains)	2–3 globular heads (500 kDa) on a common stem	Control movement along microtubules; mutants: reduced motility
Inner dynein arms (light chains)	14–120 kDa; associated with radial spokes	Primary effectors of movement; mutants: no motility
Radial links (spokes)	Protein complex	Connect each outer doublet to the central pair
Central pair and sheath	Ring-like helical sheath around the central doublet	Anchors radial links
Tektins	With outer doublets	Structural
Nexin links	Join adjacent outer doublets	Maintain cylindrical axonemal shape

- **Figure 67.22 legend** (Fawcett, 1975): plasma membrane omitted; cross-sections show acrosomal cap, connecting piece, mitochondrial sheath, annulus, fibrous sheath and endpiece in order.

Pathophysiology

- Structural defects occur in **every compartment**. Head shape abnormalities are categorised by the **strict (Kruger) classification**.
- **Most head abnormalities do not reflect sperm DNA quality** — exceptions: **headless (pinhead)** and **multiple-tailed (polyploid)** sperm.
- **Globozoospermia** (acrosomeless, round-headed): severe head phenotype; usually no DNA payload problem, but causes infertility and **failed fertilisation with IVF and even ICSI**; a recently identified cause is dysfunction of **phospholipase C zeta (PLCζ)**, the calcium-signalling oocyte-activation factor essential for fertilisation.
- **Midpiece defects** are uncommon in fertile men: **bent neck** (neck and tail at an angle $>90^\circ$), asymmetrical midpiece insertion, thick, thin or irregular midpiece.
- Tail function needs about **1000 proteins**; abnormal tail structure presents as one of two phenotypes:

CILIARY DYSKINESIA VERSUS MULTIPLE MORPHOLOGICAL ABNORMALITIES OF THE SPERM FLAGELLA

	Ciliary dyskinesia	Multiple morphological abnormalities of the sperm flagella (MMAF)
Former name	Primary ciliary dyskinesia subset: immotile cilia syndrome	Dysplasia of the fibrous sheath
Frequency	Less common	More common flagellar anomaly
Motility	Severely reduced (<10%)	Severely reduced
Morphology	Grossly normal ; concentration may be normal	Visibly absent, short, bent, coiled or irregular flagella; distortion mainly in the fibrous sheath
Ultrastructure	Axonemal and periaxonemal distortion on electron microscopy	Fibrous sheath defects
Cause	Genetic , usually recessive → normal offspring likely	Genetic (whole-exome sequencing), but also correctable causes — varicocele, excess reactive oxygen species
Associations	Primary ciliary dyskinesia: respiratory disease and dextrocardia	—

	Ciliary dyskinesia	Multiple morphological abnormalities of the sperm flagella (MMAF)
Treatment	IVF-ICSI — pregnancies and live births reported	Correct cause if possible; otherwise IVF-ICSI

Concept — *motility lives in the axoneme, so tail defects are mostly genetic and bypassed by ICSI; head defects matter chiefly where they signal a DNA problem (pinhead, polyploid) or a failure of oocyte activation (globozoospermia, PLCζ).*

KEY POINTS: SPERM

- Sperm are **ciliated cells** with a "**9 + 2**" **axoneme** that enables motility.
- Motility depends on several hundred proteins — **200–300 in the axoneme**, about 1000 across the whole tail.
- Motility defects — **ciliary dyskinesias or MMAF** — typically cause **motility <10%** and are predominantly **genetic**.
- Sperm **mtDNA** is a **circular, histone- and intron-free** ring encoding respiratory complex proteins and is **highly mutation-prone**.

SUMMARY

- **GnRH, LH and FSH** drive spermatogenesis; perturbation is a **common cause of male infertility**.
- Production is optimal **2–4 °C below body temperature**, takes about **64 days**, is **constant** through cycles and waves, and needs **very high intratesticular testosterone**.
- Sperm leave the testis **immotile and largely unable to fertilise**; epididymal transit makes them **motile and fertile**; the ejaculate **feeds, protects (antioxidant) and bars later gametes**.
- Advanced paternal age brings **genetic, epigenetic and fragmentation** changes linked to infertility and offspring disease.